Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

$\lim_{n\to\infty} \frac{n^p \sin^2(n!)}{n+1}, 0 < p < 1$ is equal to

(A) 0

(B) ∞

(C) 1

(D) None of these

Q.2

If $\lim_{x\to 0} \frac{(\sin nx)[(a-n)nx-\tan x]}{x^2} = 0$, then the value of a is:

(A) $\frac{1}{n}$

(B) $n - \frac{1}{n}$

(C) $n + \frac{1}{n}$

(D) None

Q.3

$\lim_{x\to 0} \left[\operatorname{cosec}^3 x \cdot \cot x - 2\cot^3 x \cdot \operatorname{cosec} x + \frac{\cot^4 x}{\sec x}\right]$ is equal to

(A) 1

(B) -1

(C) 0

(D) None of these

Q.4

$\lim_{x\to 0} \left(\frac{10\sin 9x}{9\sin 10x}\right)\left(\frac{8\sin 7x}{7\sin 8x}\right)\left(\frac{6\sin 5x}{5\sin 6x}\right)\left(\frac{4\sin 3x}{3\sin 4x}\right)$
$\left(\frac{\sin x}{\sin 2x}\right) =$

(A) $\frac{63}{256}$

(B) $\frac{1}{6}$

(C) $\frac{6}{5}$

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(D) $\frac{1}{2}$

Q.5

What is $\lim_{x \to 0} \frac{x}{\sqrt{1-\cos x}}$ equal to ?

(A) $\sqrt{2}$

(B) $-\sqrt{2}$

(C) $\frac{1}{\sqrt{2}}$

(D) Limit does not exist

Q.6

$\lim_{x \to \frac{\pi^-}{2}} [1 + (\cos x)^{\cos x}]^2$ is equal to

(A) does not exist

(B) 1

(C) e

(D) 4

Q.7

Let α and β be the roots of $ax^2 + bx + c = 0$
Then $\lim_{x \to \alpha} \frac{1-\cos\left(ax^2+bx+c\right)}{(x-\alpha)^2}$ is equal to :

(A) 0

(B) $\frac{1}{2}(\alpha - \beta)^2$

(C) $\frac{a^2}{2}(\alpha - \beta)^2$

(D) None of these

Q.8

If $m, n \in I_0$ and $\lim_{x \to 0} \frac{\tan 2x - n \sin x}{x^3} =$ som
integer, then value of this limit is

(A) 3

(B) 2

(C) $\frac{16+n}{12}$

(D) None of these

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Q.9

If $z_r = \cos \frac{r\alpha}{n^2} + i \sin \frac{r\alpha}{n^2}$, where $r = 1, 2, 3, \ldots, n$, then $\lim_{n\to\infty} z_1 z_2 z_3 \ldots z_n$ is equal to

(A) $\cos\alpha + i\sin\alpha$

(B) $\cos(\alpha/2) - i\sin(\alpha/2)$

(C) $e^{i\alpha/2}$

(D) $\sqrt[3]{e^{i\alpha}}$

Q.10

The value of $\lim_{x\to\infty}[\cos\sqrt{x+1} - \cos\sqrt{x}]$ is

(A) 2

(B) 1

(C) 0

(D) None of these

Q.11

The value of $\lim_{x\to\infty} x\cos\left(\frac{\pi}{4x}\right)\sin\left(\frac{\pi}{4x}\right)$ is

(A) $\frac{\pi}{2}$

(B) $\frac{\pi}{4}$

(C) 1

(D) None of these

Q.12

Let $f(x) = \begin{cases} x\sin\left(\frac{1}{x}\right) + \sin\left(\frac{1}{x^2}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$ then $\lim_{x\to\infty} f(x)$ equals

(A) 0

(B) $-1/2$

(C) 1

(D) None of these

Q.13

$\lim_{\theta\to 0} \frac{4\theta(\tan\theta - \sin\theta)}{(1-\cos 2\theta)^2}$ is

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(A) $1/\sqrt{2}$

(B) $1/2$

(C) 1

(D) 2

Q.14

The value of $\lim_{x\to\frac{1}{a}} \frac{\sin\left(cx^2+bx+a\right)}{(x-\alpha)}$ (where α and β are roots of $ax^2+bx+c=0$) is

(A) $c\left(\frac{\beta-\alpha}{\alpha\beta}\right)$

(B) $\left(\frac{\beta-\alpha}{\alpha}\right)$

(C) $\frac{1}{\alpha}$

(D) $\frac{c}{\alpha}\left(\frac{1}{\alpha}-\frac{1}{\beta}\right)$

Q.15

$\lim_{x\to 0} \frac{5^x-3^x}{\sqrt[3]{1+x}}$ is equal to

(A) $3\log 5/3$

(B) $3\log 3/5$

(C) $\log 3/5$

(D) 1

Q.16

The value of $\lim_{x\to 0} \frac{(4^x-1)^3}{\sin\frac{x^2}{4}\log(1+3x)}$, is

(A) $\frac{4}{3}(\ln 4)^2$

(B) $\frac{4}{3}(\ln 4)^3$

(C) $\frac{3}{2}(\ln 4)^2$

(D) $\frac{3}{2}(\ln 4)^3$

Q.17

$\lim_{x\to 0}(\operatorname{cosec} x)^{1/\log x}$ is equal to :

(A) 0

(B) 1

(C) $\frac{1}{e}$

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(D) None of these

Q.18

If $x > 0$ and g is a bounded function, then $\lim_{n\to\infty} \frac{f(x)e^{nx}+g(x)}{e^{nx}+1}$ is

(A) 0

(B) $f(x)$

(C) $g(x)$

(D) None of these

Q.19

The value of $\lim_{x\to 0} \frac{27^x-9^x-3^x+1}{\sqrt{2}-\sqrt{1+\cos x}}$ is

(A) $4\sqrt{2}(\log 3)^2$

(B) $8\sqrt{2}(\log 3)^2$

(C) $2\sqrt{2}(\log 3)^2$

(D) None of these

Q.20

$\lim_{x\to\pi/2} \frac{\left[\frac{x}{2}\right]}{\ln(\sin x)}$ (where [.] denotes the greatest integer function)

(A) does not exist

(B) equals 1

(C) equals 0

(D) equals -1

Answers & Solutions

Q.1 Answer:

0

Solution:

Given limit can be written as

$\lim_{n\to 0} \frac{\sin^2(n!)}{n^{1-p}(1+1/n)} (0 < p < 1)$

$= \frac{\text{some real number in } [0,1]}{\infty} = 0 \quad (\because 1-p > 0)$

Q.2 Answer:

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$n+\frac{1}{n}$

Solution:

Let $\lim_{x\to 0}\frac{(\sin nx)[(a-n)nx-\tan x]}{x^2}=0$

$\Rightarrow \lim_{x\to 0}\frac{\left(nx-\frac{n^3x^3}{3!}\right)\left[n(a-n)x-\left\{x+\frac{x^3}{3}+\ldots\ldots\right\}\right]}{x^2}=0$

(By using expansion of $\sin x$ **and** $\tan x$ **)** $\Rightarrow \quad n^2(a-n)-n=0\Rightarrow an-n^2-1=0$

$\Rightarrow a=\frac{n^2+1}{n}=n+\frac{1}{n}$

Q.3 Answer:

1

Solution:

$\lim_{h\to 0}\left[\operatorname{cosec}^3 x\cdot\cot x-2\cot^3 x\cdot\operatorname{cosec} x+\frac{\cot^4 x}{\sec x}\right.$

$=\lim_{x\to 0}\left(\frac{\cos x}{\sin^4 x}-\frac{2\cos^3 x}{\sin^4 x}+\frac{\cos^5 x}{\sin^4 x}\right)$

$=\lim_{x\to 0}\frac{\cos x\left(1-\cos^2 x\right)^2}{\sin^4 x}=\lim_{x\to 0}\cos x=1.$

Q.4 Answer:

$\frac{1}{2}$

Solution:

By using the rule, $\lim_{x\to 0}\frac{\sin x}{x}=1$

$$\text{We get, req. limit } =\frac{10}{9}\cdot\frac{9}{10}\cdot\frac{8}{7}\cdot\frac{7}{8}\cdot\frac{6}{5}\cdot\frac{5}{6}\cdot\frac{4}{3}\cdot\frac{3}{4}\cdot\frac{1}{2}$$

$$=\frac{1}{2}$$

Q.5 Answer:

Limit does not exist

Solution:

$\lim_{x\to 0}\frac{x}{\sqrt{1-\cos x}}$

$=\lim_{x\to 0}\frac{x}{\sqrt{1-\left(1-2\sin^2\frac{x}{2}\right)}}$

$=\lim_{x\to 0}\frac{x}{\sqrt{2\sin^2\frac{x}{2}}}=\frac{1}{\sqrt{2}}\lim_{x\to 0}\frac{x}{\left|\sin\frac{x}{2}\right|}$

$\text{L.H.L } =\mathrm{f}(0-h)=\lim_{\mathrm{h}\to 0}\frac{h}{\left|\sin\frac{-b}{2}\right|}$

$=-\frac{1}{\sqrt{2}}\lim_{x\to 0}\frac{2\left(\frac{h}{2}\right)}{\sin\frac{h}{2}}$

$=\frac{1}{\sqrt{2}}\times 2\times 1\quad\left(\because\lim_{\theta\to 0}\frac{\theta}{\sin\theta}=1\right)$

$=\sqrt{2}$

$\text{RHL } =f(0+h)=\lim_{h\to 0}f(0+h)$

$=\frac{1}{\sqrt{2}}\lim_{\mathrm{h}\to 0}\frac{2\left(\frac{h}{2}\right)}{\sin\frac{h}{2}}=\frac{-1}{\sqrt{2}}\times 2\times 1$

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Therefore limit does not exist.

Q.6 Answer:

4

Solution:

Given $\lim_{x\to\frac{\pi^-}{2}}$

Let $y=\lim_{x\to\frac{\pi^-}{2}}(\cos x)^{\cos x}$

$\log(y)=\lim_{x\to\frac{\pi^-}{2}}(\cos x)\log\cos x$

$\log(y)=\lim_{x\to\frac{\pi^-}{2}}\frac{\log(\cos x)}{\sec(x)}$ $\left(\frac{\infty}{\infty}\right.$ **form** $)$

Applying L'Hospital's rule $\log(y)=\lim_{x\to\frac{\pi^-}{2}}\frac{-\sin x}{\cos x(\sec x\tan x)}$

$=\lim_{x\to\pi/2^-}(-\cos x)=0$

$\therefore y=e^0=1$ **Now, limit is** $(1+1)^2=2^2=4.$

Q.7 Answer:

$\frac{a^2}{2}(\alpha-\beta)^2$

Solution:

$\lim_{x\to\alpha}\frac{1-\cos\left(ax^2+bx+c\right)}{(x-\alpha)^2}$

$=\lim_{x\to\alpha}\frac{2\sin^2\left(\frac{ax^2+bx+c}{2}\right)}{(x-\alpha)^2}$

$=\lim_{x\to\alpha}2\left[\frac{\sin\frac{a(x-\alpha)(x-\beta)}{2}}{\frac{a(x-\alpha)(x-\beta)}{2}}\right]^2\times\frac{a^2(x-\beta)^2}{4}$

$=\frac{a^2}{2}(\alpha-\beta)^2$

$\left[\right.$ **using** $ax^2+bx+c=a(x-\alpha)(x-\beta)\left.\right]$

Q.8 Answer:

3

Solution:

$\lim_{x\to0}\frac{\tan 2x-n\sin x}{x^3}=I$

$=\lim_{x\to0}\frac{2x+\frac{8x^3}{3}\ldots\ldots-nx+\frac{nx^3}{3!}}{x^3}=I$

$=\lim_{x\to0}\frac{(2-n)x+\left(\frac{16+n}{6}\right)x^3+\ldots}{x^3}=I$

$\Rightarrow n=2$ **and, thus required value** $=\frac{16+n}{6}=3.$

Q.9 Answer:

$e^{i\alpha/2}$

Solution:

$z_r=\cos\frac{r\alpha}{n^2}+i\sin\frac{r\alpha}{n^2};$

$z_1=\cos\frac{\alpha}{n^2}+i\sin\frac{\alpha}{n^2}$

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$z_2 = \cos\frac{2\alpha}{n^2} + i\sin\frac{2\alpha}{n^2}; \ldots$
$z_n = \cos\frac{n\alpha}{n^2} + i\sin\frac{n\alpha}{n^2}$
consider $\lim_{n\to\infty}(z_1 z_2 z_3 \ldots\ldots z_n)$
$= \lim_{n\to\infty}\left[\cos\left\{\frac{\alpha}{n^2}(1+2+3+\ldots.+n)\right\}\right.$
$\left.+i\sin\left\{\frac{\alpha}{n^2}(1+2+3+\ldots+n)\right\}\right.$
$= \lim_{n\to\infty}\left[\cos\frac{\alpha n(n+1)}{2n^2} + i\sin\frac{\alpha n(n+1)}{2n^2}\right]$
$= \lim_{x\to\infty}\frac{\cos\alpha(1+1/n)}{2} + \frac{i\sin\alpha(1+1/n)}{2}$
$= \cos\frac{\alpha}{2} + i\sin\frac{\alpha}{2} = e^{\frac{i\alpha}{2}}$

Q.10 Answer:

0

Solution:

$-\lim_{x\to\infty} 2\sin\frac{\sqrt{x+1}+\sqrt{x}}{2}\sin\frac{\sqrt{x+1}-\sqrt{x}}{2}$
$= -\lim_{x\to\infty} 2\sin\frac{\sqrt{x+1}+\sqrt{x}}{2} = \quad \sin\frac{\frac{\sqrt{x+1}-\sqrt{x}}{2}}{\frac{\sqrt{x+1}-\sqrt{x}}{2}}\left(\frac{\sqrt{x+1}-\sqrt{x}}{2}\right)$
$= (\textbf{ finite }) \times 1 \times \textbf{zero} = 0 \quad \left(\because \lim_{x\to\infty}\frac{\sqrt{x+1}-\sqrt{x}}{2} = 0\right)$

Q.11 Answer:

$\frac{\pi}{4}$

Solution:

$\lim_{x\to\infty} x\cos\left(\frac{\pi}{4x}\right)\sin\left(\frac{\pi}{4x}\right)$
$= \lim_{x\to\infty}\frac{x}{2}\cdot\sin\left\{2\left(\frac{\pi}{4x}\right)\right\} = \lim_{x\to\infty}\frac{\sin\left(\frac{\pi}{2x}\right)}{\left(\frac{\pi}{2x}\right)}\cdot\left(\frac{\pi}{2}\right)\cdot\frac{1}{2} = (1)\cdot\left(\frac{\pi}{2}\right)\cdot\frac{1}{2} = \frac{\pi}{4}$

Q.12 Answer:

1

Solution:

$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} x\sin\left(\frac{1}{x}\right) + \sin\left(\frac{1}{x^2}\right)$
$= \lim_{x\to\infty}\frac{\sin\left(\frac{1}{x}\right)}{\left(\frac{1}{x}\right)} + \lim_{x\to\infty}\sin\left(\frac{1}{x^2}\right) = 1+0 = 1$

Q.13 Answer:

$1/2$

Solution:

$\lim_{\theta\to 0}\frac{4\theta(\tan\theta-\sin\theta)}{(1-\cos 2\theta)^2} = \lim_{\theta\to 0}\frac{4\theta\sin\theta(1-\cos\theta)}{4\sin^4\theta\cos\theta} = \lim_{\theta\to 0}\left(\frac{\theta}{\sin\theta}\right)\frac{2\sin^2\theta/2}{\sin^2\theta\cos\theta}$
$\lim_{\theta\to 0}\frac{2\sin^2\theta/2}{(2\sin(\theta/2)\cos(\theta/2))^2}\frac{1}{\cos\theta} = \lim_{\theta\to 0}\frac{1}{2}\frac{1}{\cos^2(\theta/2)\cdot\cos\theta} = \frac{1}{2}$

Q.14 Answer:

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$\frac{c}{\alpha}\left(\frac{1}{\alpha}-\frac{1}{\beta}\right)$

Solution:

Roots of $ax^2+bx+c=0$ **are** α **and** β

so roots of $cx^2+bx+a=0$ **are** $\frac{1}{\alpha}$ **and** $\frac{1}{\beta}$

or $cx^2+bx+a=c\left(x-\frac{1}{a}\right)\left(x-\frac{1}{\beta}\right)$

Now,

$\lim_{x\to\frac{1}{\alpha}}\frac{\sin\left[c\left(x-\frac{1}{\alpha}\right)\left(x-\frac{1}{\beta}\right)\right]}{\alpha c\left(x-\frac{1}{\alpha}\right)\left(x-\frac{1}{\beta}\right)}\cdot c\left(x-\frac{1}{\beta}\right)$

$=\frac{c}{\alpha}\lim_{x\to\frac{1}{a}}\frac{\sin\left[c\left(x-\frac{1}{\alpha}\right)\left(x-\frac{1}{\beta}\right)\right]}{c\left(x-\frac{1}{\alpha}\right)\left(x-\frac{1}{\beta}\right)}\cdot\lim_{x\to\frac{1}{\alpha}}\left(x-\frac{1}{\beta}\right) \quad =$

$\frac{c}{\alpha}\left(\frac{1}{\alpha}-\frac{1}{\beta}\right) \quad \left(\because \lim_{x\to0}\frac{\sin x}{x}=1\right)$

Q.15 Answer:

$3\log 5/3$

Solution:

$\frac{0}{0}$ **form, so limit** $=\lim_{x\to0}\frac{5^x\log 5-3^x\log 3}{\frac{1}{3}(1+x)^{-2/3}}$

$=\frac{\log 5-\log 3}{\frac{1}{3}}=3\log\frac{5}{3}$

Q.16 Answer:

$\frac{4}{3}(\ln 4)^3$

Solution:

$$\lim_{x\to0}\frac{(4^x-1)^3}{\sin\frac{x^2}{4}\log(1+3x)}$$

$$=\lim_{x\to0}\frac{(4^x-1)^3}{x^3}\cdot\frac{(x/2)^2}{\sin x^2/4}\cdot\frac{3x}{\log(1+3x)}\cdot\frac{4}{3}$$

$$=\frac{4}{3}(\log_e 4)^3\cdot1\cdot\log_e(e)=\frac{4}{3}(\log_e 4)^3$$

Q.17 Answer:

$\frac{1}{e}$

Solution:

Let $y=\lim_{x\to0}(\operatorname{cosec}x)^{1/\log x}$

Taking log on both sides, we get $\log y=\lim_{x\to0}\frac{\log\operatorname{cosec}x}{\log x}$ $\left[\frac{\infty}{\infty}\right.$ **form** $\left.\right]$

$=\lim_{x\to0}\frac{-\cot x}{1/x}$ **(By L' Hospital rule)**

$=-\lim_{x\to0}\frac{x}{\tan x} \quad \left(\because \cot x=\frac{1}{\tan x}\right)$

$\Rightarrow \quad \log y=-1$

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$\Rightarrow y = e^{-1} = \frac{1}{e}$

Hence, required limit $= \frac{1}{e}$

Q.18 Answer:

$f(x)$

Solution:

Given that, $x > 0$ **and** $g(x)$ **is bounded function.**

$$\begin{aligned}\text{Limit} &= \lim_{n\to\infty} \frac{f(x)e^{nx} + g(x)}{e^{nx} + 1} \\ &= \lim_{n\to\infty} \frac{f(x)}{1 + \left(\frac{1}{e^{nx}}\right)} + \frac{g(x)}{e^{nx} + 1} = \frac{f(x)}{1+0} + \frac{\text{finite}}{\infty} \\ &= \mathrm{f(x)}\end{aligned}$$

Q.19 Answer:

$8\sqrt{2}(\log 3)^2$

Solution:

$\lim_{x\to 0} \frac{27^x - 9^x - 3^x + 1}{\sqrt{2} - \sqrt{1+\cos x}}$

$= \lim_{x\to 0} \frac{9^x \cdot 3^x - 9^x - 3^x + 1}{\sqrt{2} - \sqrt{2}\cos\frac{x}{2}}$

$= \lim_{x\to 0} \left(\frac{9^x - 1}{x}\right) \cdot \left(\frac{3^x - 1}{x}\right) \cdot \frac{1}{\sqrt{2}} \cdot x^2 \cdot \frac{1}{2\sin^2\frac{x}{4}}$

$= \lim_{x\to 0} \left(\frac{9^x - 1}{x}\right) \cdot \left(\frac{3^x - 1}{x}\right) \cdot \frac{1}{\sqrt{2}} \left(\frac{x^2/16}{\sin^2 x/4}\right) 8$

$= \frac{8}{\sqrt{2}}(\log 9)(\log 3) = 8\sqrt{2}(\log 3)^2$

Q.20 Answer:

equals 0

Solution:

$\because \frac{\pi}{4} < 1, \therefore \left[\frac{\pi}{4}\right] = 0$

$\therefore \lim_{x\to\pi/2} \frac{\left[\frac{x}{2}\right]}{\ln(\sin x)} = 0.$

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